Seasonality Anomalies in the US Equity Market

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Introduction

As Halloween approaches, investors revisit a rule of thumb old enough to have entered the language. Sell in May and go away, then return at the end of October. Bouman and Jacobsen (2002) documented the pattern and found that the November to April half-year delivered materially higher returns than the May to October half-year in most of the markets they examined. While McLean and Pontiff (2016) find that return predictability decays substantially after publication, this article asks what remains of these seasonality anomalies in the US equity market and whether trading on them has become less profitable.

The S&P 500 index proxies the US equity market, and the three-month Treasury bill secondary market rate approximates the risk-free rate. This study examines daily closing levels and their logarithmic returns r_t from 1981 to 2023.[1]

1 Seasonality Analysis

1.1 Monthly Seasonality

The Halloween strategy posits that November to April delivers materially higher returns than May to October. The January effect is the hypothesis that securities prices rise more in January than in any other month, documented for US equities by Rozeff and Kinney (1976).

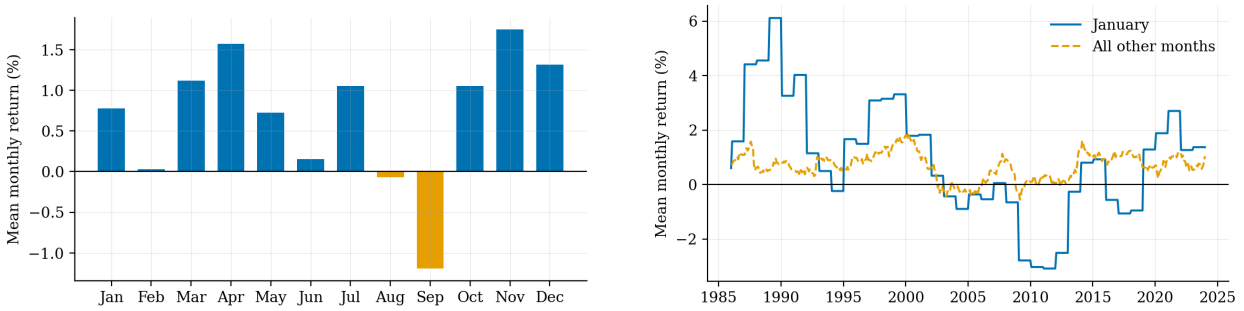


Figure 1. Mean monthly return and the 5Y rolling average of January against all other months

Figure 1 shows the mean monthly return and the five-year rolling comparison of January against all other months. January returns on average 0.77% a month, which is slightly above the sample mean. The rolling comparison changes sign in half of all windows, so the January effect is not stable within this sample.

To investigate monthly seasonality more formally, we employ a classical additive decomposition. Writing the observed return as the sum of a trend, a seasonal component and a residual, we estimate the seasonal component at each calendar position as the mean of the detrended series at that position,

$$\hat{s}_m = \frac{1}{|Y_m|} \sum_{y \in Y_m} \left(R_{y,m} - \bar{R}_y \right), \qquad m = 1, \ldots, 12, \tag{1}$$

[1] Code and replication data are available at *GitHub*.

where $\bar{R}_y$ is the mean monthly return in year y and Y_m is the set of years in which month m is observed.

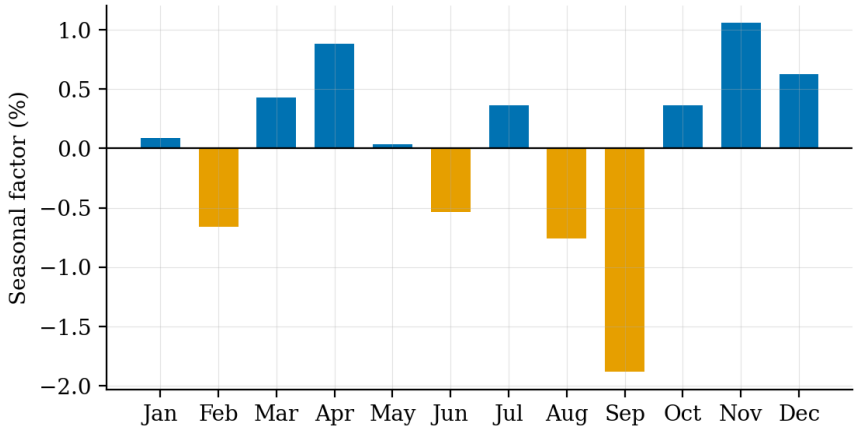


Figure 2. Estimated monthly seasonal factor

The estimated seasonal factors in Figure 2 give support to the Halloween strategy. November at +1.06% and April at +0.88% are the two strongest months, and both fall inside the November to April window, while September at −1.88% is by a wide margin the weakest and falls outside it. In this sample, the monthly pattern is primarily driven by the weakness of September and the strength of the year-end. Figure 3 compares the two half-years directly. The November to April average exceeds the May to October average by 0.81% per month. The gap is mainly positive, being negative in only 12.5% of five-year windows.

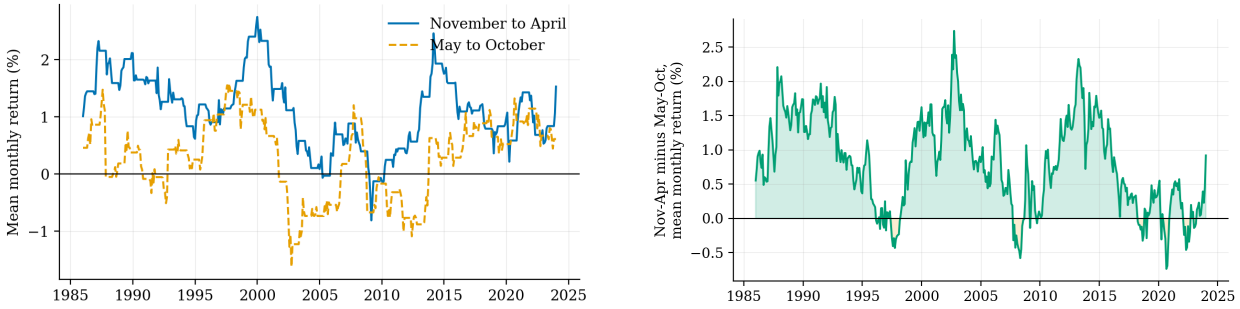


Figure 3. 5Y rolling average monthly return in each half-year, and the difference between them

1.2 Daily Seasonality

Beyond monthly effects, daily seasonality is also present. The turn-of-month effect holds that returns are abnormally high around the boundary between calendar months (Ariel, 1987). Since months contain between 15 and 23 trading days in this sample, we analyze a forward position j counted from the first trading day and a backward position counted from the last. For a month with $n_{y,m}$ trading days and daily mean $\bar{r}_{y,m}$,

$$\hat{s}_j^{+} = \operatorname*{mean}_{(y,m)} \left(r_{y,m,j} - \bar{r}_{y,m} \right), \qquad \hat{s}_j^{-} = \operatorname*{mean}_{(y,m)} \left(r_{y,m,n_{y,m}-j+1} - \bar{r}_{y,m} \right). \tag{2}$$

Figure 4 reports the seasonality of the first and last five trading days. The first trading day carries a factor of +0.147% and the second +0.061%, while the fourth and fifth are negative. At the other end of the month, although the final trading day is indistinguishable from zero at −0.010%,

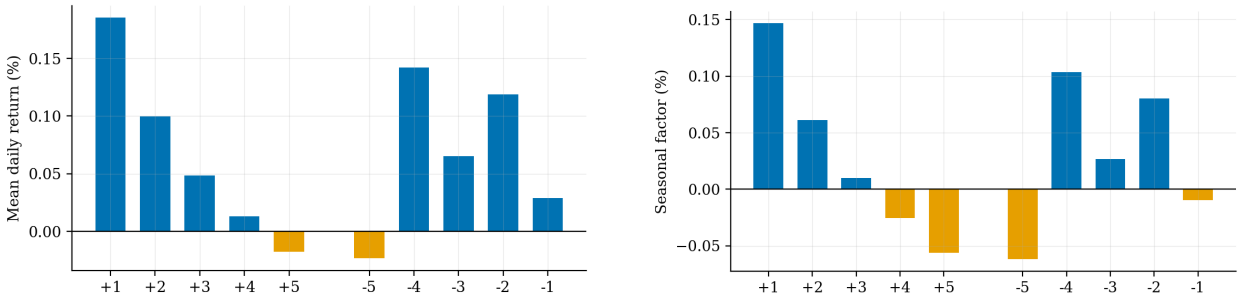


Figure 4. Mean daily return and estimated daily seasonal factor over the first and the last five trading days

the fourth, third and second to last are all positive.

1.3 January Barometer

The January barometer asserts that the January return predicts the return over the remainder of the year (Cooper et al., 2006). In our evaluation sample, it does not survive examination. The Pearson correlation between the January return and the February-to-December return is +0.156 across 43 years and is not statistically significant. Figure 5 shows that five-year rolling Pearson and Kendall correlations change sign repeatedly and show no stable relationship. Hence, the barometer is rejected.

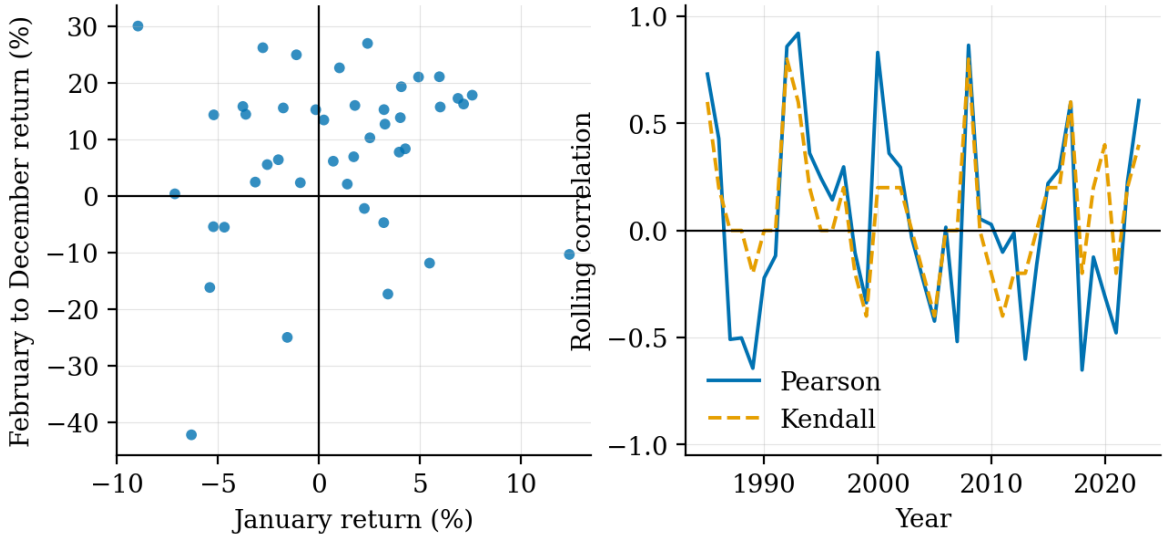


Figure 5. 5Y rolling correlation between the January return and the February to December return

2 Backtest Result and Discussion

2.1 Trading Strategies

S1 Halloween Strategy One buys at the close of the last trading day of October and sells at the close of the last trading day of April, holding Treasury bills for the remainder of the year.

S2 Halloween and Turn-of-Month Strategy On top of S1, one also holds the index over the turn-of-month window, the first and the last five trading days of each calendar month. Where the two signals coincide, the position is levered to 1.5 with the additional half funded at the bill rate.

P1 Buy and Hold One buys the index at the start of the sample and holds it throughout. Its return excludes dividends.

P2 Buy and Hold with Dividend The same position, credited with the sample average annual dividend yield of 2.48% accrued evenly across trading days.

2.2 Backtest Result

Strategy	Return (%)	Volatility (%)	Sharpe	Sortino	Max DD (%)
S1 Halloween	8.48	12.5	0.360	0.508	−37.2
S2 Halloween and turn-of-month	10.7	17.9	0.377	0.533	−48.2
P1 Buy and hold	8.27	18.1	0.238	0.328	−56.8
P2 Buy and hold with dividend	10.8	18.1	0.376	0.520	−55.2

Table 1. Performance from 1981 to 2023; returns and volatility are annualized.

The backtest charges 5 basis points (bps) one-way on every change in position. Table 1 reports the backtest performance. Both strategies outperform the P1 Buy and Hold strategy excluding dividends on every measure. However, neither exceeds the total return benchmark on return, and their Sharpe ratios sit within 0.02 of it. S2 Halloween and turn-of-month strategy has very close return, volatility and Sharpe ratio compared to P2, and its maximum drawdown is 7% shallower.

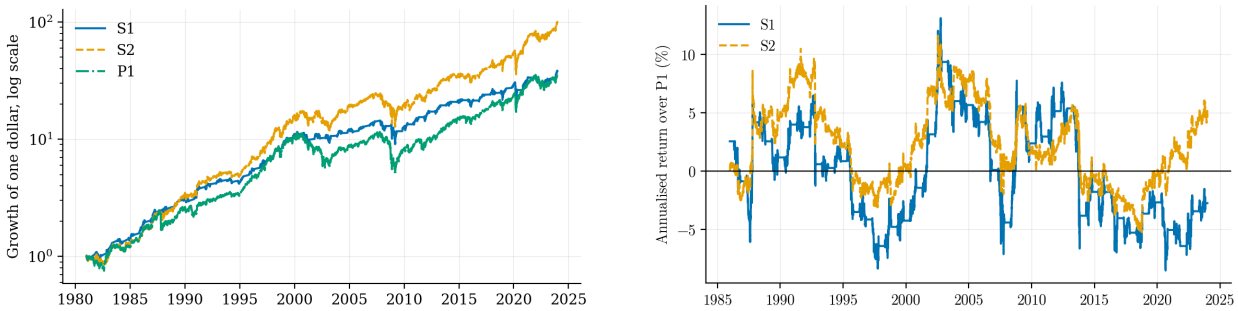


Figure 6. Growth performance under S1, S2 and P1, and the 5Y rolling outperformance over P1

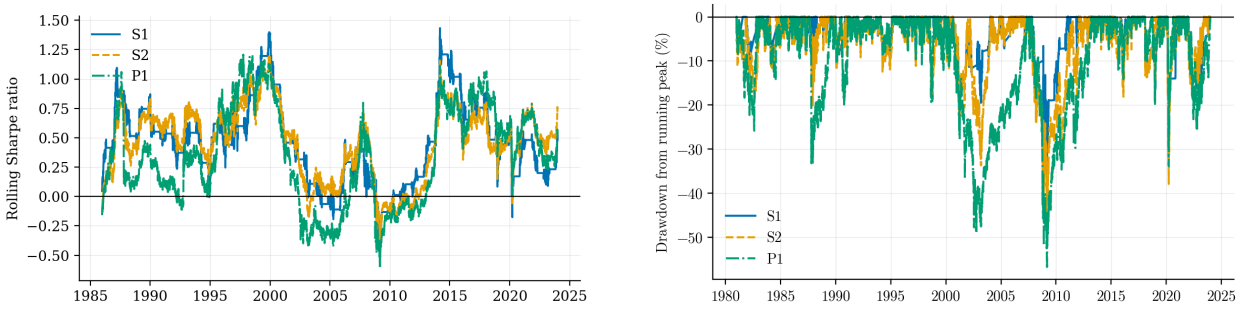


Figure 7. 5Y rolling Sharpe ratio, and drawdown from running peak

Figure 6 traces the strategies against P1 over time. The five-year rolling outperformance of S2 averages +2.38% before 2000, +2.25% from 2000 to 2019 and +2.03% from 2020. However, S2's outperformance is not statistically significant. Figure 7 turns to risk. The rolling Sharpe ratio of S2 leads P1 by 0.153 before 2000, by 0.098 from 2000 to 2019 and by 0.048 from 2020, a monotone narrowing. The drawdown panel shows S1 is shallower than P1 on 86% of days and S2 on 63%,

with the gap widest during the dot-com crisis around 2000 and the global financial crisis around 2008.

3 Discussion

In this study, we analyze monthly and daily seasonality in the US equity market, backtesting the Halloween (Sell in May and go away, then return at the end of October) strategy and the turn-of-month effect over the period of 1981 to 2023. S2 matches P2 on return, volatility and Sharpe ratio, and improves on its drawdown. We find that the rolling outperformances in return and Sharpe decrease over time and is not statistically significant.

Nevertheless, we cannot conclude that there was alpha in trading seasonality anomalies in the past, nor that there is alpha decay. Our study did not capture the effect of technological advancement. Today, every data vendor offers real-time or intraday price feeds, corporate calendars, and economic releases, whereas a hedge fund two decades ago might have waited days to receive cleaned data. Meanwhile, trading infrastructure has transformed beyond recognition. Electronic trading platforms and direct API access to exchanges have driven both commissions and latency to historic lows. An active fund in the 1980s might have paid higher commissions and waited minutes for a fill, while retail platforms today can charge zero commissions and execute orders in milliseconds. Our backtest, charging 5 bps and neglecting other frictions such as slippage, market impact and taxes, might not reflect the reality of trading in the earlier decades. Understating historical costs inflates the early-period outperformance and therefore inflates any apparent decay.

3.1 Moving Forwards

Our analysis has focused exclusively on the S&P 500, which likely understates the full extent of calendar-based anomalies. Small- and mid-cap stocks, as well as many international markets, may exhibit stronger seasonality due to lower liquidity, narrower analyst coverage, and higher access barriers. On the modeling side, our seasonal decomposition might be too naive, and a more robust framework, such as a Bayesian Structural Time Series model, could potentially estimate and forecast the seasonality factor better. Furthermore, incorporating variables such as aggregate fund flows, liquidity cycle indicators, shifts in investor sentiment, tax-loss-harvesting windows, monetary policy transitions and recessionary episodes into a multivariate setup could identify hidden patterns beyond seasonality.

References

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